

18 — Architecture Case Studies

Overview

This folder contains six complete real-world database architecture case studies modeled in PostgreSQL, plus a comprehensive interview guide. Each case study includes full schema definitions, ER diagrams, indexing strategies, query patterns, and scaling discussions.

Files

File	System	Core Theme
01_amazon_marketplace_db.md	Amazon-like Marketplace	Orders, inventory, payments, reviews, multi-seller
02_netflix_streaming_platform.md	Netflix-like Streaming	Content licensing, watch history, recommendations, billing
03_uber_ridesharing.md	Uber-like Ridesharing	PostGIS geospatial, real-time location, surge pricing
04_banking_platform.md	Banking Platform	Double-entry ledger, ACID transfers, fraud detection, compliance
05_saas_product.md	Multi-Tenant SaaS	Tenant isolation, feature flags, usage metering, RLS
06_messaging_system.md	Messaging System	LISTEN/NOTIFY, billions of messages, read receipts, presence
07_architecture_interview_guide.md	Interview Guide	30+ questions with detailed answers, key numbers, pitfalls

Key Concepts Covered

Schema Design Patterns

- Multi-tenant isolation (Row-Level Security, org_id everywhere)
- Double-entry bookkeeping (banking ledger)
- Event sourcing and audit trails
- Hierarchical data (categories with ltree)
- Polymorphic relationships
- Soft deletes vs. hard deletes

Indexing Techniques

- B-Tree composite indexes for range + equality
- GIN indexes for full-text search and JSONB
- GiST indexes for geospatial (PostGIS) and exclusion constraints
- Partial indexes for sparse conditions
- INCLUDE columns for index-only scans
- BRIN indexes for append-only time-series

Scaling Strategies Covered

Scale Threshold	Technique
> 1GB tables	Proper indexing, VACUUM tuning
> 10GB tables	Partitioning (range by date)
> 100GB / high write QPS	Read replicas, connection pooling
> 1TB	Functional decomposition (separate DBs by domain)
> 10TB	Horizontal sharding (Citus, manual)
> 100TB	Hybrid storage (PostgreSQL + Cassandra/DynamoDB)

PostgreSQL Features Used

- `PARTITION BY RANGE` — time-series data management
- `PostGIS` — geospatial queries and indexing
- `LISTEN/NOTIFY` — real-time push notifications
- `UNLOGGED TABLE` — ephemeral high-write data
- `INSERT ... ON CONFLICT` — upserts and idempotency
- `WITH RECURSIVE` — hierarchical data traversal
- `EXCLUDE USING GIST` — no-overlap constraints
- `Row-Level Security` — multi-tenant data isolation
- `GENERATED ALWAYS AS` — computed columns
- `TSVECTOR + GIN` — full-text search
- `pg_notify` — real-time events from triggers

How to Study This Material

For Schema Design Interviews

1. Read one case study end-to-end.
2. Close the document and try to draw the ER diagram from memory.
3. Write the `CREATE TABLE` statements for the 3 most important tables.
4. Identify the top 3 indexing decisions and explain why.
5. Describe at what scale each table needs partitioning.

For System Design Interviews

1. Start with `07_architecture_interview_guide.md` for the framework.
2. Study the case study closest to the system you're designing.
3. Review `21_System_Design/08_system_design_framework.md` for presentation tips.

Key Patterns to Memorize

- **Idempotency:** `INSERT ... ON CONFLICT (idempotency_key) DO NOTHING`
- **Optimistic locking:** `WHERE version = $read_version , check rowcount`
- **Consistent locking order:** Lock lower-ID row first to prevent deadlocks
- **Cursor pagination:** `WHERE created_at < $cursor ORDER BY created_at DESC`
- **Quota enforcement:** Redis counter + async PostgreSQL write
- **SKIP LOCKED:** Job queue pattern for parallel workers

Capacity Estimation Quick Reference

System	Users	Daily TXs	Annual Storage
Marketplace	300M	3M orders	~8TB hot
Streaming	250M subscribers	200M watch events	10TB+
Ridesharing	100M riders	15M trips	~5TB
Banking	10M customers	500M transactions	~10TB regulated
SaaS	5M users / 100K orgs	1B API calls	~20TB
Messaging	500M users	60B messages	4PB/yr (hybrid)

Prerequisites

- Chapter 4: Transactions and ACID
- Chapter 5: Table Partitioning
- Chapter 6: Isolation Levels
- Chapter 8: Row-Level Security
- Chapter 9: Full-Text Search
- Chapter 10: LISTEN/NOTIFY
- Chapter 11: PostGIS Extension

Next Steps

After completing this folder, proceed to:

- `21_System_Design/` — end-to-end system design documents
- Practice the questions in `07_architecture_interview_guide.md`
- Build one of these schemas locally and run the query examples